

A.4 Rigid body mechanics [HL only]

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

Show your working. $I_{\text{ring}} = mr^2$; $I_{\text{disc}} = \frac{1}{2}mr^2$.

Question 1 — The seesaw negotiation (5 marks)

A 30 kg child sits 2.0 m from the pivot of a seesaw. Her father has mass 75 kg.

- (a) Define the torque of a force about a pivot. [2]
- (b) Calculate how far from the pivot the father must sit to balance the seesaw. [2]
- (c) State the condition for a rigid body to be in rotational equilibrium. [1]

Question 2 — The pirouette (8 marks)

An ice skater spins freely with her arms out. Pulling her arms in halves her moment of inertia.

- (a) State why her angular momentum is conserved during the move. [2]
- (b) Determine the factor by which her angular velocity changes. [2]
- (c) Show that her rotational kinetic energy *doubles*, and state where the extra energy comes from. [3]
- (d) A puzzled friend argues “energy is always conserved, so the spin energy cannot change”. Point out the flaw. [1]

Question 3 — The great cheese roll (7 marks)

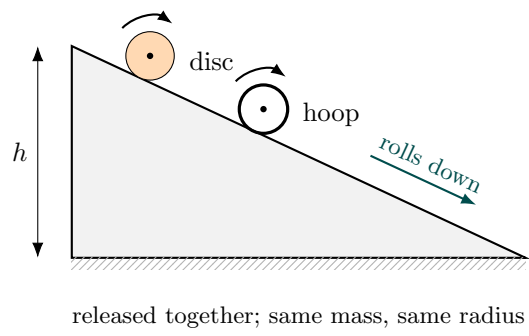
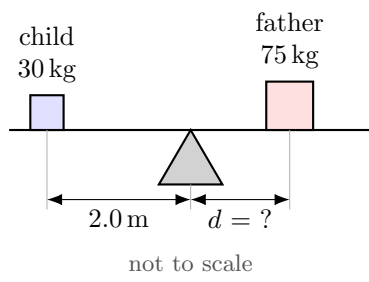
At a village fair, a solid cheese wheel (a uniform disc) and a hollow metal hoop of the *same mass and radius* are released together down a grassy slope and roll without slipping.

- (a) For the rolling disc, show that $\frac{E_{\text{rot}}}{E_{\text{trans}}} = \frac{1}{2}$. [3]
- (b) State the corresponding ratio for the hoop. [1]
- (c) Predict which object reaches the bottom first, with a one-line energy argument. [3]

Question 4 — The potter’s wheel (5 marks)

A potter’s wheel, initially at rest, accelerates uniformly to $10\pi \text{ rad s}^{-1}$ in 4.0 s.

- (a) Calculate the angular acceleration. [2]
- (b) Calculate the number of complete revolutions made during the 4.0 s. [3]



Answers

Question 1 — The seesaw negotiation

- (a) $\tau = Fr \sin \theta$: the force times the perpendicular distance from the pivot to the line of action of the force.
- (b) Balancing torques: $30 \times 9.81 \times 2.0 = 75 \times 9.81 \times d$, so $d = (30 \times 2.0)/75 = 0.80 \text{ m}$ (note g cancels from both sides).
- (c) The net torque about any point is zero. For full equilibrium the net force must be zero as well.

Question 2 — The pirouette

- (a) No external torque acts about her spin axis: the ice is effectively frictionless, and the forces her arms exert are internal to her own body.
- (b) $L = I\omega$ is constant, so halving I doubles ω .
- (c) $E = \frac{1}{2}I\omega^2$. Afterwards $\frac{1}{2} \left(\frac{I}{2}\right) (2\omega)^2 = I\omega^2 = 2 \left(\frac{1}{2}I\omega^2\right) = 2E$. The extra energy is work done by her muscles, pulling her arms inward against the outward inertial tendency of the rotating masses.
- (d) Mechanical energy is only constant when no work is done on the system. Her muscles *do* work, so the spin energy can legitimately increase.

Question 3 — The great cheese roll

- (a) With $I = \frac{1}{2}mr^2$ and $\omega = v/r$: $E_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{1}{2} \left(\frac{1}{2}mr^2\right) \left(\frac{v}{r}\right)^2 = \frac{1}{4}mv^2$. Since $E_{\text{trans}} = \frac{1}{2}mv^2$, the ratio is $\frac{1}{4}/\frac{1}{2} = \frac{1}{2}$.
- (b) Hoop: $I = mr^2$, so $E_{\text{rot}} = \frac{1}{2}mv^2 = E_{\text{trans}}$ and the ratio is 1.
- (c) The disc (the cheese). Of the shared mgh , the disc spends only $\frac{1}{3}$ on rotation against the hoop's $\frac{1}{2}$, leaving more for translation. Its translational speed is therefore greater at every height, so it reaches the bottom first.

Question 4 — The potter's wheel

- (a) $\alpha = \frac{\Delta\omega}{\Delta t} = \frac{10\pi}{4.0} = 7.9 \text{ rad s}^{-2}$.
- (b) $\theta = \frac{1}{2}(\omega_i + \omega_f)t = \frac{1}{2}(10\pi)(4.0) = 20\pi \text{ rad}$, so the number of revolutions is $20\pi/2\pi = 10$.